Phoenix Environmental & Flood Risk GIS Analysis

A screening-level spatial analysis of EPA-registered environmental facilities in Phoenix, Arizona relative to FEMA-mapped flood hazard zones.

Project Goal

Identify environmental facilities in Phoenix / Maricopa County that fall within or near FEMA flood hazard zones, in order to flag facilities that may warrant closer review of flood-related environmental risk (e.g. contaminant release during a flood event).

Data Sources

- Phoenix city boundary: US Census Bureau TIGER/Line Places, 2023 vintage.
- Environmental facilities: EPA Facility Registry Service (FRS), Arizona state-combined CSV bulk download.
- Flood hazard zones: FEMA National Flood Hazard Layer (NFHL), ArcGIS REST MapServer, Layer 28 ("Flood Hazard Zones").

Full source details, access dates, and known limitations are documented in data_sources.md.

Methodology

- Downloaded and cleaned all three source layers; validated/repaired geometries; dropped facility records with missing or invalid coordinates.
- Spatially filtered EPA facilities to those falling within the Phoenix boundary using point-in-polygon containment (not city-name text matching).
- Reprojected all layers to a projected coordinate system (EPSG:2223, NAD83 / Arizona Central State Plane, US feet) and buffered each facility by 0.5 mile.
- Spatially joined facility buffers against FEMA flood hazard polygons to flag facilities within the buffer distance of a Special Flood Hazard Area (SFHA).
- Aggregated per-facility results into summary counts by category and flood-risk status.

Coordinate Reference System (CRS) Notes

Buffering and spatial joins were performed in EPSG:2223 (a projected CRS measured in US feet), not in EPSG:4326 (geographic WGS84, measured in decimal degrees). A degree of longitude represents a different real-world distance depending on latitude, so a "0.5 degree" buffer would not equal 0.5 miles anywhere, and would be inconsistent across the study area. All layers are stored in and web-mapped from EPSG:4326, since Folium/Leaflet require geographic coordinates, but every distance calculation happens in the projected CRS.

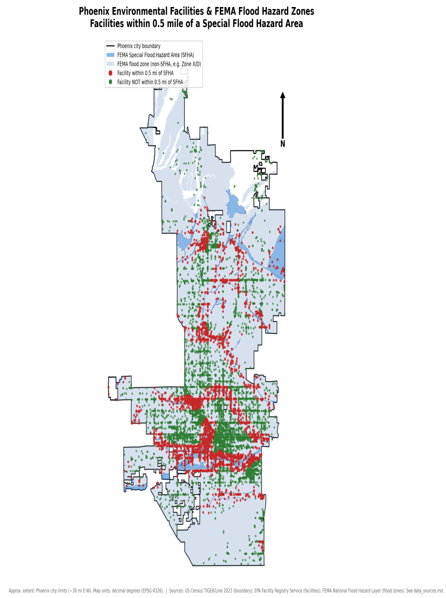
Key Findings

• 10,265 EPA-registered environmental facilities in Phoenix were analyzed.
• 4,195 facilities (40.9%) are located within 0.5 mile of a FEMA Special Flood Hazard Area (SFHA).
• The most common facility category overall is "Other / State-Registered Facility" (5,406 facilities).
• Among facilities within 0.5 mi of a FEMA SFHA, "Other / State-Registered Facility" is the most frequent category (2,241 facilities).

Facility Category Breakdown

Facility Category	Count
Other / State-Registered Facility	5,406
Hazardous Waste (RCRA)	3,537
Water Discharge (NPDES)	844
Water/Air Compliance (ICIS)	264
Air Emissions (EIS)	76
Toxics Release Inventory (TRI)	75
Air Emissions (AIRS/AFS)	37
Superfund / Contaminated Site	14
Chemical Reporting (Biennial Report)	9
Greenhouse Gas Reporting	3

Map



Full-resolution static map: maps/final_map.png. Interactive version: maps/interactive_map.html.

Data Quality Notes

EPA FRS facility records with missing, zero, or out-of-range latitude/longitude were dropped prior to analysis (see data_sources.md and clean_data.py for exact counts). A small number of FEMA flood zone polygons required geometry repair (self-intersecting rings) before use. Facility "category" is derived from the EPA program acronym(s) associated with each facility (e.g. RCRAINFO, NPDES); facilities with only minor/state program enrollments fall into an "Other / State-Registered" bucket.

Limitations

This is a screening-level proximity analysis, not a hydraulic or engineering flood risk assessment. Proximity to a mapped flood zone does not by itself indicate actual flood exposure (e.g. elevation, drainage infrastructure, and levees are not modeled). The EPA FRS facility category classification used here is a simplified relabeling of raw program-enrollment codes, not an official EPA facility-type taxonomy. FEMA flood maps are periodically revised and may not reflect the most current conditions.

Recommended Next Steps

- Incorporate facility-level hazardous substance/quantity data (e.g. TRI release volumes) to weight risk beyond simple proximity.
- Add a digital elevation model (DEM) to assess relative facility elevation within flood zones.
- Extend the analysis to all of Maricopa County using county GIS parcel/zoning data.
- Validate a sample of results in QGIS desktop against the source FEMA FIRM panels.